02YMECH - Solution 6

Assigned for the week of Oct 27, 2025

Solutions

1. **Phase–space diagrams**

   - **Left diagram:** Trajectories form closed orbits around the origin. Motion is bounded in time. The equilibrium at the origin is stable.
   - **Right diagram:** Trajectories move away from the origin. The equilibrium at the origin is *unstable*.

2. **Particle motion in polar coordinates**

   Given
   $$r(t) = r_0 + v_r t, \qquad \theta(t) = \Omega t + \tfrac{1}{2}\alpha t^2.$$

   (a) General expressions:
   $$\vec{v} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta},$$
   $$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}.$$

   (b) Components:
   $$v_r = v_r, \qquad v_\theta = (r_0 + v_r t)(\Omega + \alpha t),$$
   $$a_r = -(r_0 + v_r t)(\Omega + \alpha t)^2,$$
   $$a_\theta = (r_0 + v_r t)\alpha + 2v_r(\Omega + \alpha t).$$

   (c) Speed and its derivative:
   $$v(t) = \sqrt{v_r^2 + (r_0 + v_r t)^2(\Omega + \alpha t)^2},$$
   $$\dot{v} = \frac{(r_0 + v_r t)(\Omega + \alpha t)\,[\alpha(r_0 + v_r t) + v_r(\Omega + \alpha t)]}{v(t)}.$$

   The sign of $\dot{v}$ is determined by the numerator.

   (d) Radial acceleration vanishes when
   $$a_r = 0 \;\Rightarrow\; t^* = -\frac{r_0}{v_r} \quad \text{or} \quad t^* = -\frac{\Omega}{\alpha}.$$

   Such times exist for $t^* \geq 0$ if $v_r < 0$ and/or $\Omega\alpha < 0$.